CROW CARRIER — eight synchronous microphone channels / USB data / 12 V operating power

PROTOTYPE — NOT FOR FABRICATION. Outdoor protection and exact assembly qualification remain open.

12 V INPUT / REVERSE PROTECTION / 5 V CONVERSION
J1
PJ-102AH / 12V CENTER POSITIVE
F1
0451001.MRL 1A
FUSED12
D3
SS34-E3/57T
12V
U10
R-78E5.0-0.5
5V
D2
Diodes Inc 1N5819HW-7-F
VSYS
D1
Littelfuse SMBJ15A
C1
10uF 35V X7R 20%
C2
10uF 35V X7R 20%
PWR1
PWR2
PWR3
PWR4
PWR_FLAG
U3 ASSEMBLY: DEPOPULATE ON-MODULE D1 only; retain R10/R1 VBUS sense. USB DATA ONLY.
U3
Pico SC0915 MOD; on-module D1 DEPOPULATED
SHARED 12.288 MHz MCLK
X1
CB3LV-3I-12M2880000
MCLK_SRC
R1
4.7k 1%
R2
4.7k 1%
R3
68R 1% 0.1W 25V
MCLK_FAN
R4
56R 1% 0.1W 25V
MCLK_ADC1
R5
56R 1% 0.1W 25V
MCLK_ADC2
ADC 1: CHANNELS 1–4 / PCM1864 / common 3V3
U1
PCM1864DBT
U1 LOCAL DECOUPLING — separate capacitor positions at each supply pin
ADC 2: CHANNELS 5–8 / PCM1864 / common 3V3
U2
PCM1864DBT
U2 LOCAL DECOUPLING — separate capacitor positions at each supply pin
POD 1 — differential audio + power + active-low calibration command
POD 2 — differential audio + power + active-low calibration command
POD 3 — differential audio + power + active-low calibration command
POD 4 — differential audio + power + active-low calibration command
POD 5 — differential audio + power + active-low calibration command
POD 6 — differential audio + power + active-low calibration command
POD 7 — differential audio + power + active-low calibration command
POD 8 — differential audio + power + active-low calibration command
RJHSE-5080
SP0504BAHTG
2.2uF 25V X7R 20%
2.7k 1%
BZT52C3V9-7-F
2.2nF 50V C0G 5%
100k 1%
DMN3023L-7
1206L010/30WR

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